TERRASCOPE — DISASTER DAMAGE INTELLIGENCE

Official Automated Building Damage Assessment Report | Powered by xView2 1st Place Engine

Assessment ID: asm-1db4190d53	**Disaster Type:** Earthquake
Assessment Name: test20	**Event Date:** 2026-10-12
Location: None	**Created At:** 2026-10-12T15:44:51

1. Damage Summary & Metrics

Metric	Value	Description
Total Detected Buildings	**194**	Distinct building structures identified in localized area
Assessed Structure Footprint	**16121.0 sq.m**	Estimated footprint coverage derived from satellite resolution
Model Execution Duration	**18.84 seconds**	Inference runtime using PyTorch CPU pipeline

2. Damage Severity Classification Breakdown

Damage Category	Count	Percentage	Hex Indicator
No Damage	61	31.4%	#16A34A (Green)
Minor Damage	104	53.6%	#CA8A04 (Yellow)
Major Damage	25	12.9%	#EA580C (Orange)
Destroyed	4	2.1%	#DC2626 (Red)

3. Spatially Aligned Damage Map Visualization



4. Technical Engine & Methodological Notes

Inference Model: xView2 First-Place Solution (Siam-ResNet34 U-Net)
Checkpoints: xView2 Siamese ResNet34 U-Net (PyTorch CPU Engine)
Execution Engine: PyTorch CPU Multi-threaded Matrix Operations
Pipeline: Dual-stage inference combining building localization mask with 4-class Siamese feature concatenation.

Notice & Operational Disclaimer: *This assessment report is produced automatically by the TerraScope computer vision damage inference engine based on satellite image pairs. Automated predictions should serve as preliminary decision support for emergency response teams and must be verified through ground-truth inspections or secondary GIS verification prior to final tactical deployment.*